Retina

Figures: 8.20, 8.1, 8.2, 8.4, 8.8, 8.9, 15.12, 8.32, 8.12, 8.10, 8.13, 8.27, 8.23, 8.22, 8.25, 8.26, 8.28, 8.30, 8.31

I. General retinal anatomy

A. Terms

1. Macula and fovea

2. Posterior pole

3. Peripheral retina

4. Fundus

B. Overview of retinal pathway

1. Photoreceptors

2. Bipolar cells

3. Ganglion cells

II. Retinal histology

A. Retinal pigment epithelium (RPE)

1. Single cell thick

a. Longer, narrower, and more densely pigmented in the macular area

2. The inner portion of the RPE, consists of microvilli that extend into the photoreceptor layer.

3. Functions of the RPE

a. Blood-retinal barrier

b. Phagocytosis of photoreceptor discs

c. Metabolizes and stores vitamin A

d. Produces growth factors

(1) Secretes vascular endothelial growth factor (VEGF) which helps maintain choriocapillaris function

e. Absorbs light

B. Photoreceptor cells (rods and cones)

1. Contain photopigments that are activated by light

2. Composition of rods and cones:

a. Outer segment

b. Cilium

c. Inner segment

(1) Two parts

i. Ellipsoid:

ii. Myoid:

d. Outer fiber

e. Cell body

f. Inner fiber

3. Photoreceptors secrete the neurotransmitter glutamate

C. Bipolar cells

1. Dendrite synapses with
2. Axon synapses with
    a. Secrete the neurotransmitter glutamate

D. Ganglion cells

1. Dendrite synapses with

2. Axon synapses with

3. Classification:

    a. P-cells terminate in the parvocellular layers of the LGN

    b. M-cells aka parasol ganglion cells project to the magnocellular layers of the LGN

    c. Intrinsically photosensitive ganglion cells (ipRGC)

E. Horizontal cells

1. Synapse with

2. Function

    a. Role in visual integration

    b. Modulate cone responses

F. Amacrine cells

1. Synapses with

2. Function

    a. Modulate the information that reaches the ganglion cells

G. Muller cells

1. Extend through much of the retina

2. Function

    a. Provide structure to the retina

    b. Absorbs ions, neurotransmitters, and metabolic waste products

    c. Metabolize, synthesize, and store glycogen

III. Ten retinal layers

A. Alternate grouping of neurons and their processes

1. Retinal pigment epithelium (RPE)

2. Photoreceptor layer

3. External limiting membrane (ELM)

4. Outer nuclear layer (ONL)

5. Outer plexiform layer (OPL)

6. Inner nuclear layer (INL)

7. Inner plexiform layer (IPL)

8. Ganglion cell layer (GCL)

9. Nerve fiber layer (NFL)

    a. Clinical correlate: Retinal hemorrhages

10. Internal limiting membrane (ILM)

    a. Clinical correlate: Retinal sheen

IV. Retinal regions

A. Macula (macula lutea)

1. Darkened region in the central retina

2. Choroid is thicker than other areas of the retina

3. May have a yellow hue due to higher concentration xanthophyll pigments (lutein and zeaxanthin) than other areas of the retina.

a. Function

B. Fovea (fovea centralis)

1. The bipolar and ganglion cells are displaced laterally

2. 1 mm inferior to the center of the disc

3. Rod free zone

4. Capillary-free zone

C. Ora serrata

1. Peripheral termination of the retina

2. 5 mm anterior to the equator

3. Firm attachment between the retina and vitreous

D. Optic disc aka optic nerve head

1. Ganglion cell axons accumulate and exit the eye

2. The central retinal artery enters and the central retinal vein exits the eye

3. Dimensions

    a. 1.7 mm horizontally
    b. 1.9 mm vertically

4. Physiologic cup

V. Retinal blood supply

B. Outer retinal layers (photoreceptors)

C. Inner retinal layers

D. Three capillary networks

1. Radial peripapillary capillary plexus

2. Superficial capillary plexus

3. Deep capillary plexus

4. Outer to the outer plexiform layer is avascular.

5. Outer plexiform layer receives nutrients from both retinal and choroidal circulation.

E. Cilioretinal artery

1. 15-20% of the population

2. Enters retina from the temporal edge of the disc

3. Blood supply from choroidal circulation

4. Clinical correlate: Central retinal artery occlusion